

Adjoint-Faithful Fourier Neural Operator for a Baroclinic Double Gyre

A minimum-data summer internship project plan

Core idea: first establish a frozen, adapted Bire-architecture baseline on the same MITGCM trajectories used by AF-FNO. Then improve the operator with modern forward losses and a small set of forward perturbation-response pairs. Generate true MITGCM adjoints only after the emulators are frozen, and use them as a blind test.

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Ten-week scope; official MITGCM baroclinic-gyre tutorial; NeuralOperator 2.0 implementation

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1 Recommendation in one page

Recommended path. Build an **adjoint-faithful FNO (AF-FNO)** on the official MITGCM baroclinic double-gyre tutorial at 1° . Before changing the architecture or loss, train a frozen **A0 adapted Bire FNO** on exactly the same trajectories. This tests transfer of the paper’s qualitative trends—useful short forecasts, stable rollouts, and sensitivity to wind amplitude—without claiming a bitwise reproduction of its undocumented 0.25° dataset. Then introduce the modern A–C ladder and forward-only response supervision. Do not put a single MITGCM adjoint in training or model selection.

The minimum credible programme is:

- **Three long production regimes:** low, control, and high wind amplitude.
- **Two short held-out wind runs:** the intermediate amplitudes used by Bire et al.; inference-only and not 20 years long.
- **One frozen architecture-transfer baseline (A0):** the recovered Bire FNO core, adapted only where the 62×62 , 46-state benchmark forces a change. It is trained and evaluated before the modern A–C models.
- **224 ten-day perturbed restarts:** 128 training, 32 validation, and 64 inference. They provide local tangent information from forward MITGCM only and total about 6.1 model-years.
- **Twelve true-adjoint runs:** three smooth objectives, two horizons, and two unseen initial states, opened only after the model and criteria are frozen.

This separates four questions: Does the Bire architecture transfer to the documented tutorial data? Does modernization improve the forward emulator? Did response supervision improve the local input–output map? Does automatic differentiation through it yield a useful adjoint? The defensible claim is deliberately limited: *adjoint fidelity over specified horizons, objectives, states, and a tested control subspace.*

No-adjoint-training contract. MITGCM adjoints are excluded from training, hyperparameter tuning, early stopping, loss selection, perturbation-amplitude selection, and architecture selection. The test is run only after checksums of the frozen model and evaluation script are recorded.

2 Research question and hypotheses

Let the MITGCM time- Δt map be

$$x_{n+1} = \mathcal{M}_{\Delta t}(x_n, u_n),$$

where x is the state and u contains forcing and controls. The FNO $\mathcal{F}_\theta$ approximates the same map. For a scalar objective $J(x_N)$,

$$\lambda_n^{\text{FNO}} = \left(\frac{\partial \mathcal{F}_\theta}{\partial x_n} \right)^T \lambda_{n+1}^{\text{FNO}}, \quad \lambda_N = \nabla_{x_N} J.$$

Primary question. Can local derivative information obtained exclusively from short forward perturbation experiments make these gradients agree with MITGCM adjoints more closely than a state-accurate forward FNO?

Hypotheses: (1) the adapted A0 baseline reproduces the qualitative Bire forecast and wind-response trends on the coarser tutorial; (2) a modern FNO improves forward and boundary behavior relative to A0; (3) state accuracy alone does not ensure adjoint accuracy; and (4) a local response loss improves gradient direction without adjoint labels or loss of forward skill.

3 Why local perturbation–response information is the right approach

3.1 A small state error can hide a wrong derivative

Suppose $\mathcal{M}(x) = x$, but an emulator learns

$$\mathcal{F}(x) = x + \delta \sin(kx), \quad \mathcal{F}'(x) = 1 + \delta k \cos(kx).$$

The state error never exceeds δ , yet the derivative error can be large when k is large. A value-only loss controls the height of a learned curve at sampled points; it does not adequately control its slope. An adjoint is built from those slopes.

Intuitive ocean example

Two emulators predict sea-surface height ten days ahead to within one centimetre. In one, a small extra wind-stress patch strengthens the western boundary current; in the other, it weakens it. Their forecast scores can be almost identical, but a gradient-based optimization steps in opposite directions. One perturbed forward run reveals the physical response sign without computing an adjoint.

3.2 Forward responses teach Jacobian actions

At an anchor $z = (x, u)$ and normalized direction v , run the model from z and $z + \varepsilon v$:

$$r_M(z; v, \varepsilon) = \frac{\mathcal{M}_{\Delta t}(z + \varepsilon v) - \mathcal{M}_{\Delta t}(z)}{\varepsilon} \approx D\mathcal{M}_{\Delta t}(z)v.$$

Match it with the emulator secant

$$r_\theta(z; v, \varepsilon) = \frac{\mathcal{F}_\theta(z + \varepsilon v) - \mathcal{F}_\theta(z)}{\varepsilon}.$$

This constrains a Jacobian–vector product (JVP) using only forward simulations and ordinary first-order training backpropagation. The link to adjoints is duality:

$$\underbrace{v^T D\mathcal{M}(z)^T q}_{\text{gradient projected on } v} = \underbrace{q^T D\mathcal{M}(z)v}_{\text{objective response along } v}.$$

Thus, matching $D\mathcal{M}v$ for relevant directions also matches projections of adjoints for many later objectives q . The full Jacobian remains too large to identify; the claim must stay within the sampled smooth wind and state subspace.

4 Benchmark choice: trend, not hidden dataset

The public `oceanfourcast` repository is a conceptual starting point, not an exact reproduction package. Its training defaults differ from the paper, its README points to external data, and its loader predicts

Table 1: Published setup, official tutorial, and proposed benchmark.

	Bire et al.	Official tutorial	Proposed
Grid	0.25°, 248 × 248	1°, 60 × 60 ocean cells plus land border	Tutorial first; fine grid only as stretch
Vertical	15 levels, about 2 km	15 levels, 1.8 km	Tutorial levels
Time step	300 s	1200 s	Tutorial value
τ_0 (N m ⁻²)	0.075, 0.0875, 0.100, 0.1125, 0.125	Default 0.100	Same five amplitudes
Training winds	Low, control, high	Not applicable	Low, control, high
Held-out winds	Two intermediates	Not applicable	Same two; short runs
Goal	Forward emulation	Numerical tutorial	Forward and local-adjoint fidelity

ten selected channels including diagnostic pressure and streamfunction. Reimplement the method in the maintained NeuralOperator library rather than preserve old dependencies [4, 5].

This removes uncertainty about an unavailable initialization and pipeline, reduces the horizontal cell count by about 16, and retains the relevant baroclinic physics. The tutorial notes that the deep ocean is still adjusting after 100 years; stationarity must therefore be measured for the variables and horizons used [3].

4.1 A0: a time-boxed Bire-architecture transfer baseline

Do not make the incomplete public repository a prerequisite for the project. Audit the paper, repository, and recovered implementation once; record unresolved discrepancies; then freeze A0 and label it *adapted* rather than exact. Implement it through the public NeuralOperator 2.0 **SpectralConv** interface while retaining the recovered paper structure: channel-diagonal Fourier multipliers, explicit pointwise channel mixing, three Fourier blocks, width 128, alternating sine/cosine position fields, and paper-style one-step state loss and optimizer schedule.

All models must use the same S0–S2 source states, chronological splits, masks, and training-only normalization. Only adaptations forced by the documented benchmark are allowed in A0:

Table 2: Declared adaptations from the Bire configuration to A0.

Element	Bire / recovered setting	A0 on the tutorial
Horizontal grid	248 × 248 at 0.25°	62 × 62 including the land rim
Dynamic state	Ten selected prognostic/diagnostic outputs	All 46 Markov-state channels; wind is the additional input
Fourier modes	[64, 64]	[16, 16], preserving the approximate mode-to-grid ratio
Coarse pretraining	Stride 8, yielding 31 × 31	Stride 2, also yielding 31 × 31
Forecast target	Normalized next state	Retain for A0; Models A–C predict a normalized residual
Software	Historical <code>oceanfourcast</code> environment	Recovered architecture on pinned NeuralOperator 2.0

Run a 20–100-sample overfit test, one ten-day development training, and the locked low/control/high evaluation. Do not tune A0 after opening held-out I1/I2 results. If the recovered baseline cannot learn the small sample after the declared learning-rate fallbacks, report that result and proceed; repairing

historical code must not delay the AF–FNO experiment. Separate five- and 30-day A0 models are optional follow-ups only when needed to compare a specific Bire lead-time panel.

5 Minimum MITgcm simulation programme

5.1 Five trajectory jobs; three long production regimes

Table 3: Minimum trajectories. Durations are planning budgets; diagnostics decide whether equilibration is adequate.

ID	τ_0	Duration	Initialization	Use	Output
S0	0.100	100 y spin-up; final 10 y production	Tutorial initial state	Control data; branch checkpoint	Annual restart; daily production
S1	0.075	5 y adjust + 10 y production	S0 checkpoint	Low-wind data	Daily production
S2	0.125	5 y adjust + 10 y production	S0 checkpoint	High-wind data	Daily production
I1	0.0875	1 y	Unseen S0-derived state	Interpolation inference only	Daily
I2	0.1125	1 y	Different unseen state	Interpolation inference only	Daily
Total		142 model-years	Three long production regimes; two short inference regimes		

After year 50, inspect five-year trends of heat content, surface kinetic energy, basin-integrated transport, and level-mean temperature. Stop at year 100 only if the upper-ocean quantities are statistically stationary; otherwise extend the run or narrow the first claim. The 1° grid and four-times-larger time step should be much cheaper per model-year than Bire’s setup, but wall-clock/model-year must be benchmarked.

5.2 Chronological data splits

Use years 1–7 for training, year 8 for validation, and years 9–10 for inference within each ten-year S0–S2 production segment. Insert 90-day gaps at boundaries.

The 224 perturbed restarts add 2,240 model-days, about 6.1 model-years. Save only initial/final states, configuration hash, and perturbation definition. A 46-channel 62×62 float32 snapshot is about 0.7 MB; 30 daily production years are of order 8 GB before compression.

5.3 Eight directions per response anchor

Pilot standardized amplitudes 10^{-3} , 3×10^{-3} , and 10^{-2} . Select the largest for which a central-difference linearity check is within 5–10%, then freeze the rule by direction family. Use randomly signed one-sided perturbations for training and central differences for the blind response test.

6 State, forcing, and preprocessing

Table 4: Data roles. Pair counts are approximate for daily output and a ten-day target.

Dataset	Source	Approx. size	Permitted use
Trajectory training	S0–S2 years 1–7	7,500 pairs	Fit weights
Trajectory validation	S0–S2 year 8	1,000 pairs	Tune and early-stop
Known-wind inference	S0–S2 years 9–10	2,100 pairs	Final scores only
Unseen-wind inference	I1 and I2	700 pairs	Final interpolation only
Response training	16 anchors $\times$ 8 directions	128 endpoints	Fit response loss
Response validation	4 anchors $\times$ 8 directions	32 endpoints	Tune loss and ε
Response inference	4 anchors $\times$ 8 directions $\times$ 2 signs	64 endpoints	Blind central differences
True adjoints	3 costs $\times$ 2 horizons $\times$ 2 states	12 integrations	Blind comparison only

Table 5: Minimum smooth perturbation basis.

Count	Direction family	Purpose
1	Prescribed cosine-wind amplitude	Directly tests the Bire control axis
1	Smooth meridional wind phase/shape mode	Separates amplitude from pattern learning
2	Smooth localized wind envelopes	Probe western/southern and northern/interior control
4	Balanced, low-wavenumber state directions	Training-only EOFs or filtered anomalies; rotate among anchors

6.1 Predict a Markov state, not independent diagnostics

Place the prognostic variables on a documented common tracer grid and stack vertical levels as channels:

$$x = [U_{1:15}, V_{1:15}, \Theta_{1:15}, \eta], \quad C_{\text{out}} = 46.$$

Vertical velocity, pressure, and streamfunction remain diagnostics computed by a fixed, differentiable postprocessor when needed. If salinity is active and nonconstant, add its 15 levels rather than silently omitting it.

Inputs comprise the 46 dynamic channels, wind stress, and normalized longitude, latitude, wet mask, and distance-to-wall fields. Record every C-grid interpolation operator. Compare gradients only after applying the exact transpose of those operators so the FNO and MITGCM sensitivities share a native control basis.

Compute channel means and scales from trajectory training only, mask land, and store the affine transform with the model. Gradient comparisons must undo normalization:

$$\tilde{x}_i = (x_i - \mu_i) / \sigma_i \implies \frac{\partial J}{\partial x_i} = \frac{1}{\sigma_i} \frac{\partial J}{\partial \tilde{x}_i}.$$

Models A–C predict the normalized ten-day residual $\Delta \tilde{x}_n = \tilde{x}_{n+1} - \tilde{x}_n$. This gives the network an explicit identity path and focuses capacity on change. A0 alone retains the recovered paper-style normalized next-state target; this intentional difference belongs to the architecture-transfer baseline and must be stated in every A0 comparison.

7 Modern adjoint-faithful FNO

7.1 Starting implementation

After freezing A0, use the maintained `neuraloperator` package and pin its release and commit. Begin the common A–C ladder with a modest dense FNO:

Table 6: Starting AF–FNO configuration. Validation uses only trajectories and forward responses.

Choice	Starting value	Reason / limited search
Forecast interval	10 days	Economical response runs; useful scale in Bire et al.
Fourier modes	[16, 16]	Compare 12, 16, 24 after inspecting spectra; respect Nyquist
Hidden channels	32	Increase to 64 only if a 20–100-sample overfit test fails
FNO layers	4	Current library default; compact enough for ablation
Local branch	<code>conv_bias_kernel=3</code>	Complements global mixing near boundaries
Boundary treatment	10% domain padding	Reduces periodic FFT wrap in a closed basin
Geometry	Grid embedding, mask, wall distance	Makes solid boundaries explicit
Channel MLP	Enabled	Mixes the vertical state channels locally
Precision	float32	Avoid mixed precision in the initial derivative study
Factorization	Dense FNO first	TFNO is a memory fallback, not an initial confounder

Select mode count from the resolved power spectrum: too few modes oversmooth the western boundary current; too many learn aliased fluctuations. Domain padding or Fourier continuation is relevant because the basin is not periodic. These choices follow the NeuralOperator 2.0 practical guide rather than the outdated oceanfourcast environment [2].

7.2 Baseline ladder and controlled losses

For a one-step pair and response direction, use

$$\mathcal{L} = \mathcal{L}_{\text{state}} + \lambda_{\text{roll}}\mathcal{L}_{\text{roll}} + \lambda_{\text{resp}}\mathcal{L}_{\text{resp}} + \lambda_{\text{spec}}\mathcal{L}_{\text{spec}} + \lambda_{\text{bnd}}\mathcal{L}_{\text{bnd}}, \quad (1)$$

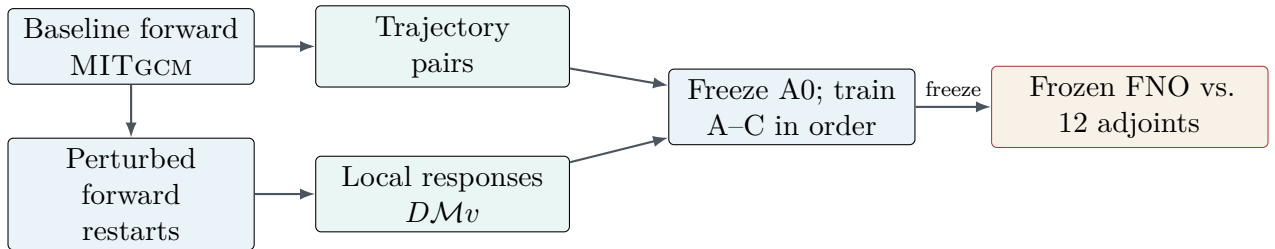
$$\mathcal{L}_{\text{resp}} = \frac{\|W_r(r_\theta - r_M)\|_2^2}{\|W_r r_M\|_2^2 + \epsilon_0}. \quad (2)$$

$\mathcal{L}_{\text{state}}$ is masked relative L_2 ; $\mathcal{L}_{\text{roll}}$ is a three-step (30-day) unrolled loss; $\mathcal{L}_{\text{spec}}$ compares binned spectra; and $\mathcal{L}_{\text{bnd}}$ upweights wet cells near the western wall. An H^1 -style spatial loss may supplement the spectral term, but it is not a substitute for $\mathcal{L}_{\text{resp}}$: spatial output derivatives and derivatives of the input–output operator are different objects.

Table 7: Required architecture-transfer baseline and ablation models.

Model	Training signal	Purpose	Expected result
A0	Adapted Bire architecture and paper-style one-step loss	Test architecture transfer before modernization	Qualitative Bire trend on the common data
A	Modern dense FNO + one-step residual state loss	Controlled modern baseline	Comparable state skill; derivatives may be poor
B	Model A + rollout + spectral/boundary losses	Isolate modern forward losses from response data	Better rollout and boundary state skill
C	Model B + local response loss	Test the AF–FNO hypothesis	Similar state skill; better responses and adjoints

Freeze A0 before training A. Tune the A–C terms incrementally. Increase λ_{resp} until validation response error improves without worsening ten-day state error by more than 5%. Lock all weights before viewing any true adjoint.



The adjoint archive has no path into training or tuning.

Figure 1: The common trajectory pipeline feeds A0 and A–C; response data enter only C, and the adjoint evaluation remains sealed.

8 Ground-truth MITgcm adjoint suite

8.1 Minimum: twelve integrations

One adjoint integration returns sensitivities to the full initial state and active controls for one scalar objective. It is therefore more efficient to vary objective, horizon, and state than to run an adjoint for every perturbation direction.

Table 8: Blind matrix: 3 objectives $\times$ 2 horizons $\times$ 2 initial states = 12 adjoint integrations.

Smooth scalar objective	Reason	Horizons	States
Gaussian interior sea-surface-height average	Broad low-wavenumber response away from walls	10, 30 days	Two unseen dates
Smooth western-boundary sea-surface-height average	Energetic region where spectral/boundary errors matter	10, 30 days	Same dates
Area-weighted barotropic gyre-strength functional	Dynamical circulation measure; avoids nondifferentiable maxima	10, 30 days	Same dates

Select dates and masks before opening any adjoint result; place dates at least 90 days from split boundaries and one another. If mandatory tests succeed, an optional six-run extension adds 90-day horizons. Before accepting a reference adjoint, run a Taylor remainder or central directional finite-difference check at one test state per objective family. This verifies MITGCM but may not alter the emulator.

8.2 Metrics and an operational test

For gradients g_M and g_F , report

$$\text{cosine} = \frac{g_F^T g_M}{\|g_F\|_2 \|g_M\|_2}, \quad \text{angle} = \cos^{-1}(\text{cosine}), \quad \text{relative error} = \frac{\|g_F - g_M\|_2}{\|g_M\|_2}.$$

Convert both gradients to the same physical units and grid with explicit area/volume weights. Report full-state scores and projections onto the directions in Table 5. Direction is primary: optimization can tolerate imperfect magnitude but not a gradient pointing uphill. Also show signed maps, vertical profiles, wavenumber spectra, and sign agreement in the largest-sensitivity region.

Finally, use the emulator gradient to propose a small control update, choose the step from a predeclared short line search, and evaluate the objective with one forward MITGCM run. This tests usefulness without turning adjoints into labels.

9 Evaluation and predeclared success criteria

Evaluate A0 beside persistence, climatology, and Models A–C on the forward tasks. Use one development seed throughout and three seeds, if affordable, for the final A versus C derivative comparison. A0 is frozen before I1/I2 are opened and is not tuned with response or adjoint data. Required plots are:

- A0/A–C RMSE and correlation versus lead against persistence and daily climatology;
- A0/A–C one-year rollout stability and spectra at low/control/high wind;
- A0/A–C skill and monotone physical response at the two unseen intermediate winds;
- response error and cosine by direction family and region;
- adjoint angle and relative error for all 12 tests, paired A versus C;

- one forward MITGCM control-update test per objective family.

Predeclared targets, not promises, are:

- **A0 transfer:** learn the small-sample map, beat persistence or climatology at useful short leads, and reproduce the direction of the wind-amplitude trend. Exact Bire error values are neither expected nor claimed.
- **Modern forward trend:** A or B improves on A0, remains stable for one year, and varies monotonically with held-out wind amplitude.
- **Response:** C lowers blind response error by at least 25% versus A while ten-day state error rises by no more than 5%.
- **Adjoint:** C improves median projected-gradient cosine by at least 0.2. Useful targets are median cosine above 0.8 at ten days and 0.6 at 30 days.
- **Usefulness:** at least 80% of predeclared emulator-gradient control steps lower the objective in forward MITGCM.

Failure remains diagnostic. Better one-step JVPs but poor 30-day adjoints implicate Jacobian composition; no JVP improvement implicates perturbation scale or subspace; interior-only success implicates boundary representation.

10 Ten-week execution plan

Table 9: Long MITGCM runs begin immediately and continue alongside code development.

Week	Activity	Output	Decision gate
1	Reproduce tutorial; benchmark; launch S0	Versioned setup, restart test, wall-clock/model-year	Runs are reproducible
2	Complete S0–S2; preprocess state maps	Common data store, masks, transpose-checked operators	Training trajectories pass QC
3	Audit and freeze A0; train ten-day baseline	Architecture manifest, small-sample overfit, A0 checkpoint	Adapted Bire map learns signal
4	Train modern Model A; spectrum study	A0/A forward comparison and baselines	Qualitative Bire trend reproduced
5	Train B; run I1/I2 unseen winds	Rollout, spectrum, and interpolation plots	Modernization helps without leakage
6	Perturbation pilot and response data	Linearity plots; training/validation responses	Perturbations resolved and local
7	Train C; final forward/response seeds	A–C ablation; sealed checkpoints and scripts	Freeze models and criteria
8	Generate and verify 12 adjoints	Taylor checks; sealed adjoint archive	Open only after freeze
9	Gradient comparison and control updates	Tables, maps, true objective changes	Optional 90-day test decision

Week	Activity	Output	Decision gate
10	Robustness, report, and handoff	Reproducible repository and manifest	Complete minimum claim

11 Risks and fallbacks

Table 10: Safeguards preserve a useful ten-week result.

Risk	Early diagnostic	Fallback
Spin-up cost or deep drift	Trends after year 50; measured cost	Scope first claim to upper ocean or use a documented equilibrated restart
Bire reconstruction ambiguity	Architecture audit exposes paper/repository conflicts	Freeze and label A0 as adapted; time-box historical repair
46-channel memory pressure	100-sample overfit and GPU profile	Reduce width, then TFNO, then vertical representation
FFT wraps across walls	Seam and boundary-response errors	More padding; Fourier continuation; local convolution and wall distance
Perturbation noise/nonlinearity	Central-difference pilot	Change ε by family before freezing it
Good ten-day, poor 30-day gradient	Horizon-separated metrics	Strengthen three-step training; report validated horizon honestly
Adjoint build delayed	Week-1 compilation smoke test	Deliver complete forward-response result; adjoint remains follow-on
Sampled directions too narrow	Good projected but poor full-map score	Add directions from response residuals as a labeled extension

The 0.25° configuration is a stretch goal only after C passes at 1° . A valuable extension is resolution transfer at 0.5° or 0.25° , testing both values and local responses. It must not displace the mandatory adjoint comparison.

12 Deliverables and reproducibility

The internship should leave:

1. version-controlled MITGCM configurations for S0–S2, I1–I2, perturbed restarts, and the 12 adjoint cases;
2. a manifest with checksums, units, grids, masks, split dates, perturbation vectors, and configuration hashes;
3. a frozen A0 architecture manifest and pinned NeuralOperator 2.0 pipelines for A0 and Models A–C;
4. separate sealed archives for forward training data and comparison-only adjoints;
5. scripts for forward, response, adjoint, and control-update metrics;
6. a report stating the validated state, subspace, horizons, objectives, and failures.

Record seeds, package lockfile, model checksum, normalization statistics, optimizer state, splits, and timing. Unit tests should cover staggered-grid interpolation and its transpose, gradient denormalization, masking, response construction, and a toy autodiff/adjoint identity.

13 Final scope statement

This project is feasible because it makes both the architecture-transfer and adjoint questions small and testable. It does not rebuild every undocumented detail of Bire et al.; A0 preserves the recoverable architecture and labels every forced adaptation, then the A–C ladder tests modernization and derivative supervision on a documented MITGCM case. The project spends new simulation budget on 224 short, information-rich perturbations instead of more long trajectories, then reserves 12 expensive adjoints for a blind comparison.

The contribution is an empirical test of a transferable principle: *local forward responses can constrain a neural operator’s derivatives enough to improve adjoints for unseen objectives, without adjoint supervision.* The baseline, ablations, sealed evaluation, and explicit subspace claim make the outcome useful whether the hypothesis succeeds or fails.

References

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